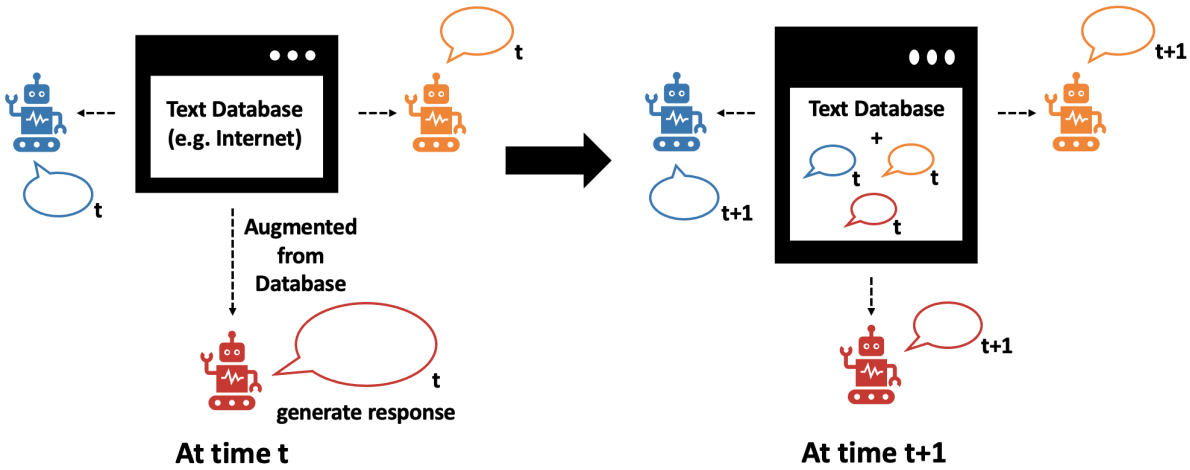

Figure 1: Experiment in Section 3.1, from t to $t+1$. At time t, three different LLMs generate responses based on augmentation contexts fetched from a shared text database. They then post their responses to the text database, based on which they generate the next-time responses.

fined as

$$D_{ij}^{(t)} := \left\| X_i^{(t)} - X_j^{(t)} \right\|_2, \; i,j = 1, \cdots, n \quad (2)$$

where $\|\cdot\|_2$ denotes the Euclidean norm. To discuss that model diversity shrinks under our experimental setting, we conjecture that $\left\|D^{(t)}\right\|_F$ converges to a small nonnegative number c in probability, i.e.

$$\left\|D^{(t)}\right\|_F \xrightarrow{p} c \text{ as } t \to \infty,$$

where $\|\cdot\|_F$ denotes the Frobenius norm, and c depends on some unknown intrinsic differences between models that are invariant to the environment. The detailed experiment is summarized in Algorithm 1, and we present examples in Section 4 to illustrate this conjecture.

3.2 A System of Gaussian Mixture Models

Next we design a system of Gaussian mixture models (GMMs) analogous to the mechanism in Section 3.1. Consider a collection of n time-varying GMMs in d-dimensional Euclidean space. For all models, we fix the number of components B, a single nondegenerate covariance matrix $\Sigma \in \mathbb{R}^{d\times d}$, and mean vectors $\mu_b \in \mathbb{R}^d$, $b = 1, \cdots, B$. We define the i-th GMM at t, denoted as $g_i^{(t)}$, via the distribution

$$g_i^{(t)}(x) = \sum_{b=1}^{B} \pi_{i,b}^{(t)} \mathcal{N}(x; \mu_b, \Sigma), \quad (3)$$

where $\mathcal{N}(x; \mu_b, \Sigma)$ denotes the density of a normal distribution at x with parameters (μ_b, Σ), and $\pi_{i,b}^{(t)}$ denotes the GMM's mixture weight at component b; we collect the B mixture weights as the probability vector $\Pi_i^{(t)}$. We initialize each GMM with different vectors $\Pi_i^{(t=0)}$. For example, one can sample each vector $\Pi_i^{(0)}$ from a Dirichlet distribution with parameter $(a, \ldots, a)$.

Recall that $\phi \circ (F_i^{(t)}(q))$ denotes the i-th LLM model at time t, evaluated by query q and embedded by model ϕ. It follows some distribution, which can be approximated by certain GMM like $g_i^{(t)}$. The parameters in $g_i^{(t)}$, including $\{B, \Sigma, \mu_b, \Pi_i^{(t)}\}$, can be all time-varying; for simplicity, we fix most of them and only allow $\Pi_i^{(t)}$ to update at each time. Hence, GMM i at time t is characterized by its mixture weights, the pairwise distance between $g_i^{(t)}$ and $g_j^{(t)}$ is reduced to the distance between $\Pi_i^{(t)}$ and $\Pi_j^{(t)}$, and we can define the $n \times n$ distance matrix at time t as $D^{(t)}$:

$$(D^{(t)})_{ij} := \left\| \Pi_i^{(t)} - \Pi_j^{(t)} \right\|_2. \quad (4)$$

We then define a set of points A, similar to the set of texts ("the Internet") in Section 3.1. Let $A^{(0)}$ be the initialization of this set. At each t, each GMM $g_i^{(t)}$ generates L points: $s_{i,1}^{(t)}, \cdots, s_{i,L}^{(t)}$ to be added to the set A:

$$A^{(t+1)} = A^{(t)} \cup \bigcup_{i=1}^{n} \{s_{i,1}^{(t)}, \cdots, s_{i,L}^{(t)}\}.$$

This set is used to update each GMM in the following way: each model i randomly draws k_t

Algorithm 1 Experiment: synthetic conversations on the Internet

Require: n different LLMs, `nomic-embed-v1.5`, a set of texts $A^{(0)}$, a fixed query q, a fixed L, a fixed β, total time T.

for timestep $t = 0$ to T **do**
  for each model $i = 1$ to n **do**
    Retrieve k_t sentences uniformly from $A^{(t)}$, where $k_t := \lfloor \beta \cdot |A^{(t)}| \rfloor$.
    Generate answers based on the retrievals L times: $(f_i^{(t)}(q))_1, \cdots, (f_i^{(t)}(q))_L$.
    Embed them: $(\phi \circ (f_i^{(t)}(q))_1, \cdots, (\phi \circ (f_i^{(t)}(q))_L$ using `nomic-embed-v1.5`.
    Compute $X_i^{(t)}$ using Eq.(1).
  end for
  Construct matrix $D^{(t)}$ using Eq.(2). **Compute** $\|D^{(t)}\|_F$.
  Update the set A: $A^{(t+1)} = A^{(t)} \cup \{(f_1^{(t)}(q))_1, \cdots, (f_n^{(t)}(q))_1\}$.
  Get $F_i^{(t+1)}(q)$ corresponding to the updated $A^{(t+1)}$, $i = 1, \cdots, n$.
end for

points from $A^{(t)}$. As in Section 3.1, we put $k_t = \lfloor \beta \cdot |A^{(t)}| \rfloor$. Using $\Pi_i^{(t)}$ as the prior probabilities and these k_t points as observed data, we compute a posterior update of $\Pi_i^{(t+1)}$ via a revised update algorithm in Zivkovic and van der Heijden (2004), as summarized in Algorithm 2:

Algorithm 2 Mixture-weight update algorithm

Require: Weight vector $\Pi_i^{(t)} = (\pi_{i,1}^{(t)}, \cdots, \pi_{i,B}^{(t)})$, k_t points $\{u_1, \cdots, u_{k_t}\}$, series of coefficients $\alpha_t \in (0, 1)$ and $c_{t,B}$, threshold $\epsilon \in [0, 1]$.

Initialize $\pi_{i,b}^{(t+1)} \leftarrow \pi_{i,b}^{(t)}$ for all $b = 1, \ldots, B$.
for u in $\{u_1, \cdots, u_{k_t}\}$ **do**
  Compute the ownership $o_b = \pi_{i,b}^{(t+1)} \mathcal{N}(u; \mu_b, \Sigma) / g_i^{(t+1)}(u)$, $b = 1, \cdots, B$; $g_i^{(t+1)}$ defined in Eq.(3).
  for $b = 1$ to B **do**
    $\pi_{i,b}^{(t+1)} \leftarrow (1 - \alpha_t)\pi_{i,b}^{(t+1)} + \alpha_t \frac{o_b - c_{t,B}}{1 - B \cdot c_{t,B}}$.
  end for
  for $b = 1$ to B **do**
    if $\pi_{i,b}^{(t+1)} \leq 0$ **then**
      for $b' \neq b$ **do**
        $\pi_{i,b'}^{(t+1)} \leftarrow \frac{\pi_{i,b'}^{(t+1)}}{1 + \epsilon - \pi_{i,b}^{(t+1)}}$.
      end for
      $\pi_{i,b}^{(t+1)} \leftarrow \epsilon$.
    end if
  end for
end for

The update of $\Pi_i^{(t)}$ thus updates $g_i^{(t)}$ to $g_i^{(t+1)}$. This experiment is summarized in Algorithm 3. For this system of GMMs, we conjecture that:

$$\|D^{(t)}\|_F \xrightarrow{p} 0 \text{ as } t \to \infty, \tag{5}$$

Section 4 presents simulation results supporting this conjecture, and we prove it in Appendix A.

4 Results

4.1 LLM Experiment

To implement the experiment designed in Section 3.1, we try $n = 3$ models: Meta's `Llama-3.1-8B-Instruct` (Dubey et al., 2024), DeepSeek's `deepseek-llm-7b-chat` (Wang et al., 2023), and Mistral's `Mistral-7B-Instruct-v0.3` (Jiang et al., 2023). These three AI companies are headquartered in the U.S., China, and France respectively; we choose them so that the three models are pre-trained differently at least due to the distinct emphasis on contexts in their own languages. At $t = 0$, we construct the "Internet" $A^{(0)}$ by putting 20 posts about cryptocurrencies, collected and organized by Ebrahimi (2024): 7 are positive, 7 are neutral, and 6 are negative. The query we use is *"Please provide EXACTLY one concise sentence about the future prospects of Bitcoin."*, and each model answers the question after reading posts sampled from $A^{(t)}$. We instruct them to answer exactly in one sentence because we want to focus on the semantic similarity or difference between these three models over time, instead of the language structure and complexity difference. The latter is also interesting to explore, but requires other statistical tools on texts, such as the depth of

Algorithm 3 Simulation of a GMM system

Require: Number of models n, dimension d, number of components B, covariance matrix Σ, mean vectors $\{\mu_b\}_{b=1}^{B}$, positive integer L, fixed coefficient β, total time T.

Initialize mixture coefficients $\Pi_i^{(t)} \sim \text{Dir}(a)$ and $A^{(0)}$.
for timestep $t = 0$ to T **do**
  Construct matrix $D^{(t)}$ using Eq.(4). **Compute** $\|D^{(t)}\|_F$.
  for each model $i = 1$ to n **do**
    Generate L points $s_{i,1}^{(t)}, \cdots, s_{i,L}^{(t)}$.
    Sample $k_t := \lfloor \beta \cdot |A^{(t)}| \rfloor$ points uniformly from $A^{(t)}$, say, $\{u_1^{(t)}, u_2^{(t)}, \ldots, u_{k_t}^{(t)}\}$.
    Update $\Pi_i^{(t)}$ to $\Pi_i^{(t+1)}$ via Algorithm 2 using data $\{u_1^{(t)}, u_2^{(t)}, \ldots, u_{k_t}^{(t)}\}$. This also updates the GMM from $g_i^{(t)}$ to $g_i^{(t+1)}$.
  end for
  Update A: $A^{(t+1)} = A^{(t)} \cup \bigcup_{i=1}^{n} \{s_{i,1}^{(t)}, \cdots, s_{i,L}^{(t)}\}$.
end for

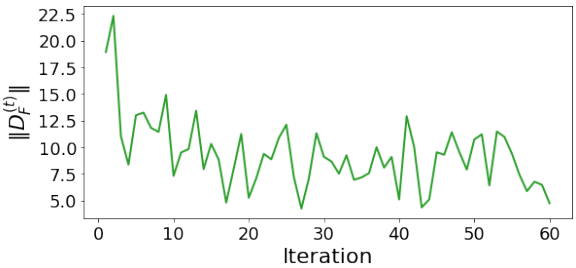

Figure 2: As t approaches 60, the norm of the distance matrix in our LLM experiment, i.e. $\left\|D^{(t)}\right\|_F$ as defined in Eq.(2), decreases from approximately 20 to around 5.

parse tree of sentences. We aim to investigate this in our future work.

We then follow Algorithm 1 to simulate the experiment for $T = 60$. We set $L = 40$ and $\beta = 0.5$. The experiment was conducted on `Nvidia A100 80GB GPUs`, `AMD EPYC 7443 24 Core`, taking approximately 8 hours to complete the full process. Figure 2 shows that the norm decreases from over 20 to around 5, indicating that the average responses generated by these three models are becoming more similar semantically.

In fact, this phenomenon is evident when directly comparing posts generated by the three models at the initial time and at $T = 60$ (see Appendix B). We can see that the models initially express opinions from different perspectives, but by $T = 60$, their outputs become nearly identical semantically.

Recall that at each time t, we generate $L = 40$ responses for each model so that we can examine their distributions. Since each embedded response is a vector of length 768, we perform Classical Multidimensional Scaling on the pairwise distance matrix so that we can plot each response in a 2-D space. From Figure 3, we can see that at the beginning, the distributions of three models' responses look very different, but at T, they are all point masses that are close to each other. This indicates that the three models are not only less diverse from each other, but also exhibit reduced internal diversity within themselves in the simulated Internet environment populated with synthetic posts.

4.2 GMM Experiment

Next we present some results of our GMM experiment described in Section 3.2 to demonstrate how it's phenomenally similar to the LLM experiment. First, we define three two-component GMMs ($B = 2$). The component means are $\mu_1 = -5$, $\mu_2 = 5$, and the variances are both $\Sigma_1 = \Sigma_2 = 1$. The initial weights are sampled from Dirichlet with $a = 1$, and same as the LLM experiment, we set $\beta = 0.5$. As suggested by Zivkovic and van der Heijden (2004), we set $\alpha_t = k_t^{-1}$; since $B \cdot c_{t,B}$ should be smaller than 1, we take $c_{t,B} = 1/(k_t \cdot B^2)$. We take $L = 3$ and $T = 200$.

Figure 4 shows how $\left\|D^{(t)}\right\|_F$ evolves in 5 replicates (i.e. by running the simulation 5 times with different seeds) - they all seem to converge to 0.

Since GMM has an analytical form of density, we plot histograms of our three GMMs at different times. From Figure 5, we can see that they start with very different component weights but seem to converge to a same weight vector.

Indeed, we prove that for $i, j \in \{1, \cdots, n\}$, $\Pi_i^{(t)} - \Pi_j^{(t)}$ converges to 0 in expectation. From this